
AI, Human Cognition and Knowledge Collapse

Acemoglu, Kong & Ozdaglar (2026)

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github.com/carlosgomezp28/ai-04-acemoglu

Paper and the agent's problem

PAPER RESULT

Human learning creates private context-specific information and general knowledge for future cohorts; agentic AI directly supplies context-specific precision.

Information and production

$$Y = \sigma^{-2} + \lambda_I e + \tau_A.$$

Assumption 1

$$\Delta_I = 0, \quad \Delta_X > 0.$$

$$\Delta_I = f(0, 1) - f(0, 0).$$

Thus context-specific knowledge alone has no incremental productive value.

Static choice of human effort

$$\max_{e \geq 0} \left\{ f(0, 0) + \Delta_G G(X) + \Delta_X G(X) G(Y) - \frac{e^\alpha}{\alpha} \right\}.$$

$$\Delta_X G(X) \lambda_I g(Y) = e^{\alpha-1}$$

X	general-knowledge precision
e	human learning effort
τ_A	agentic-AI precision
Y	total context-specific precision

Observation 1: complements and substitutes

PAPER RESULT

General knowledge complements effort

$$U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0$$

Strict when

$$\Delta_X > 0, \lambda_I > 0, X > 0, Y > 0, \\ g(X) > 0, g(Y) > 0.$$

$X \uparrow \Rightarrow$ marginal return to effort $\uparrow$.

Agentic AI substitutes for effort

$$U_{e\tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0$$

Strict when

$$\Delta_X > 0, \lambda_I > 0, X > 0, Y > 0, \\ G(X) > 0, g'(Y) < 0.$$

$\tau_A \uparrow \Rightarrow$ marginal return to effort $\downarrow$.

Welfare trap. Holding X fixed, more accurate AI raises static payoff; by crowding out learning and future general knowledge, it need not raise long-run welfare monotonically.

PAPER RESULT

Paper reproduction

- ▶ reconstructed the static objective and FOC;
- ▶ derived Observation 1 and checked boundaries;
- ▶ audited all three Section 5 extensions;
- ▶ isolated the maintained production restriction

$$\Delta_I = 0;$$

- ▶ compiled algebraic sign checks in `lean/Main.lean`.

OUR PROPOSED EXTENSION — NOT IN THE PAPER

$$\Delta_I > 0, \quad \Delta_X > 0, \quad \Delta_G = 1 - \Delta_I - \Delta_X \geq 0.$$

$$[\Delta_I + \Delta_X G(X)] \lambda_I g(Y) = e^{\alpha-1}$$

Boundary. The paper has $X = 0 \Rightarrow e^* = 0$. In our extension,

$$\Delta_I > 0, \lambda_I > 0, \alpha > 1, \sigma^{-2}, \tau_A < \infty \Rightarrow e^*(0, \tau_A) > 0.$$

Public knowledge. Given this positive boundary effort,

$$\lambda_G > 0, I > 0, 0 \leq \Sigma^2 < \infty \Rightarrow F_{\Delta_I}(0) > 0.$$

Exact zero public knowledge is no longer a fixed point for finite AI precision.

Where I did not believe AI

6. Implication for public knowledge

The baseline transition law is

$$X_{t+1} = [(X_t + \lambda_G I e^*)^{-1} + \varepsilon^2]^{-1}$$

At $X=0$

$$F_{\Delta_I}(0) = [(\lambda_G I e^*(0, \tau_A))^{-1} + \varepsilon^2]^{-1} > 0$$

provided

$$\lambda_G > 0, \quad I > 0$$

7. Final result

$$\Delta_I > 0 \rightarrow F_{\Delta_I}(0) > 0$$

Result. Positive standalone value of context-specific knowledge removes the exact zero-knowledge fixed point for finite AI precision

It doesn't imply that knowledge collapse disappears in general

OUR PROPOSED EXTENSION — NOT IN THE PAPER

INITIAL AI CLAIM

$\Delta_I > 0 \Rightarrow$ knowledge collapse is eliminated.

TOO STRONG.

AUDITED VERDICT

$$\Delta_I > 0 \Rightarrow F_{\Delta_I}(0) > 0$$

Exact zero disappears, but low positive states, multiplicity, path dependence, and welfare loss may remain.

Section 5.2: synthetic data adds exogenous public precision; ours works through endogenous human effort.